

2328482
2329678

MCTA 3371

AUTONOMOUS MOBILE ROBOT NAVIGATION

A Hybrid Fuzzy-Genetic Algorithm Approach

START

BACKGROUND

**Mobile robot navigation
requires safely moving from a
starting point to a designated
goal while successfully avoiding
static obstacles.**



SETUP

MCTA 3371

Simulation Workspace: A discrete 2D grid environment structured at 20×20 grid units.

Robot Representation: Modeled as a simple point robot moving in discrete steps (Forward, Backward, Left, Right).



SETUP

MCTA 3371

Approach A: Standard Fuzzy Logic (Reactive)

- **Uses local heuristic decision-making based on immediate surrounding inputs.**
- **Moves toward the direction with the highest score relative to the goal.**

Limitation: It has no memory or global layout awareness, making it highly vulnerable to U-shaped traps (local minima).



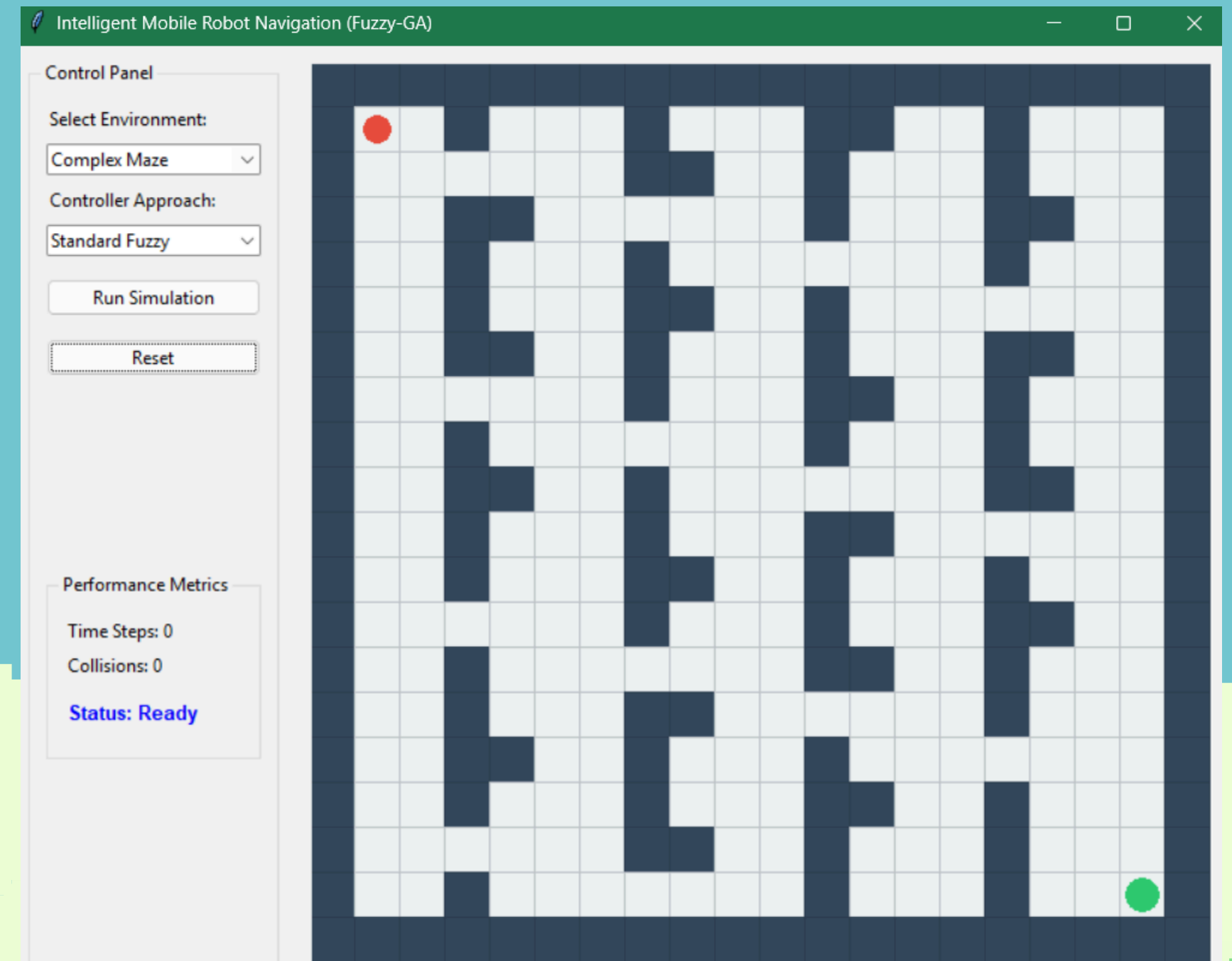
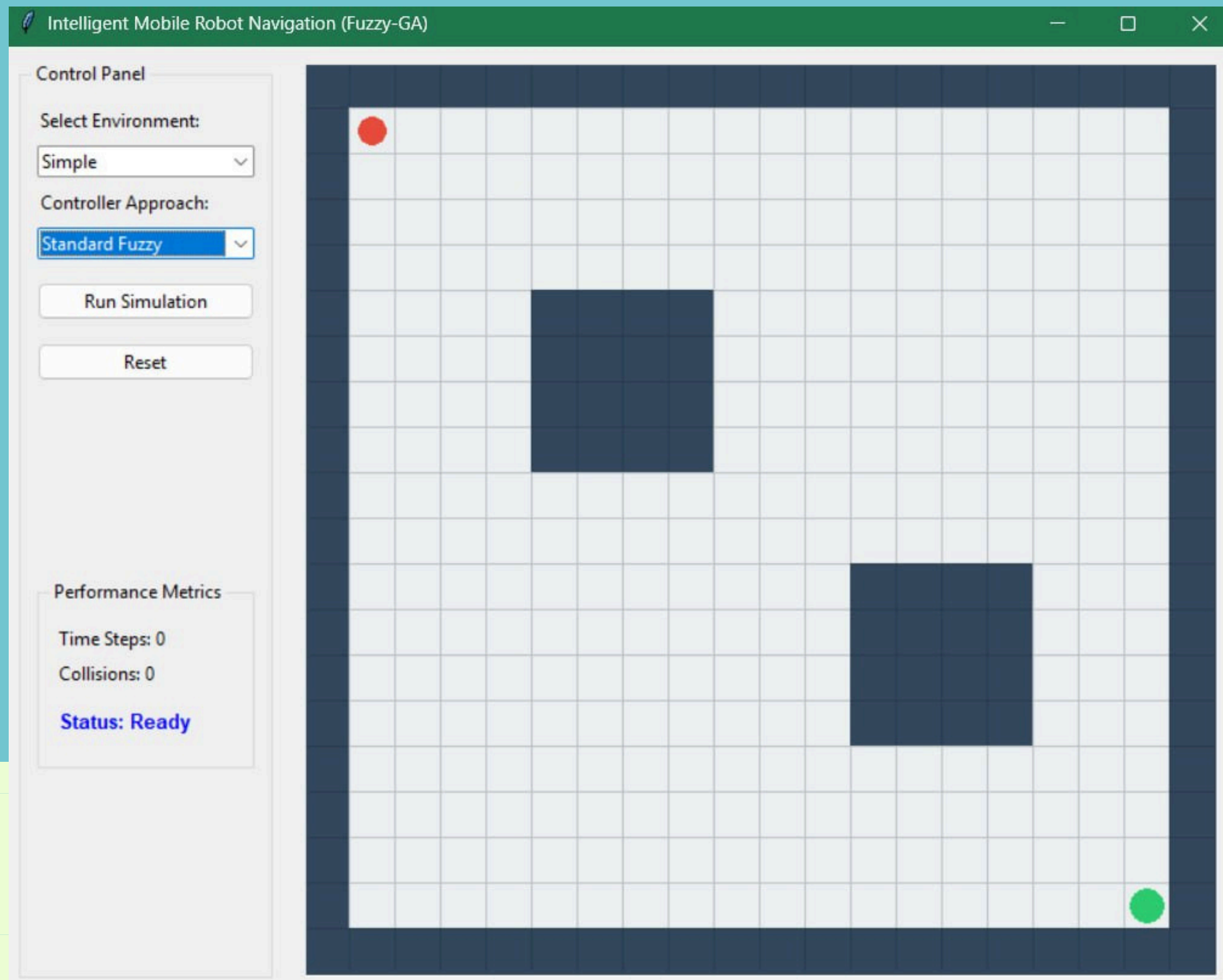
SETUP

MCTA 3371

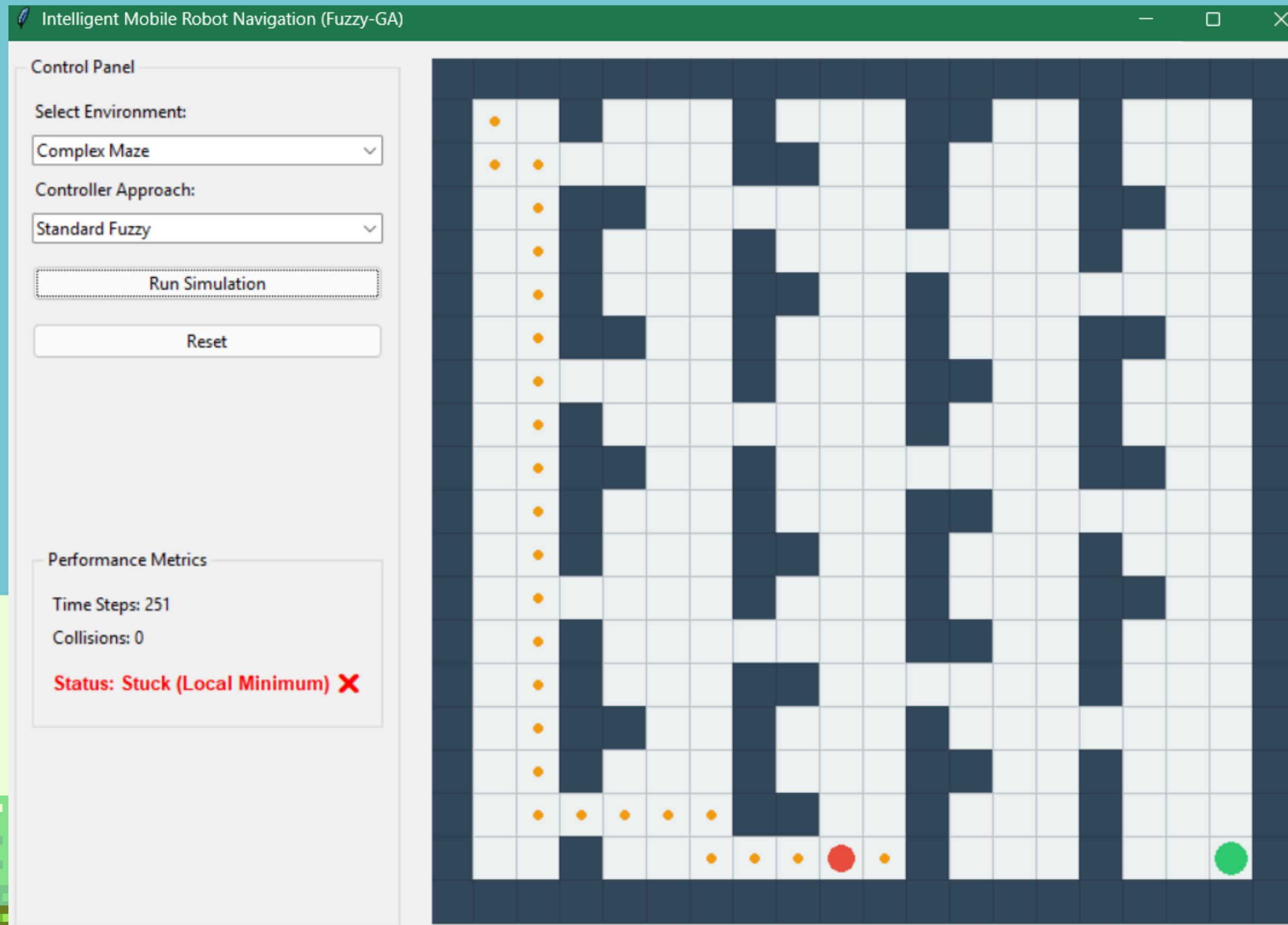
Approach B: Hybrid Fuzzy-GA (Preferred Optimization)

- Integrates a global distance-field heuristic optimized via Genetic Algorithm mechanics.
- Evaluates the entire layout to pre-calculate global path paths.
- Feeds this optimized distance array into the fuzzy framework so the robot knows when to move "away" from a wall to escape a dead end.

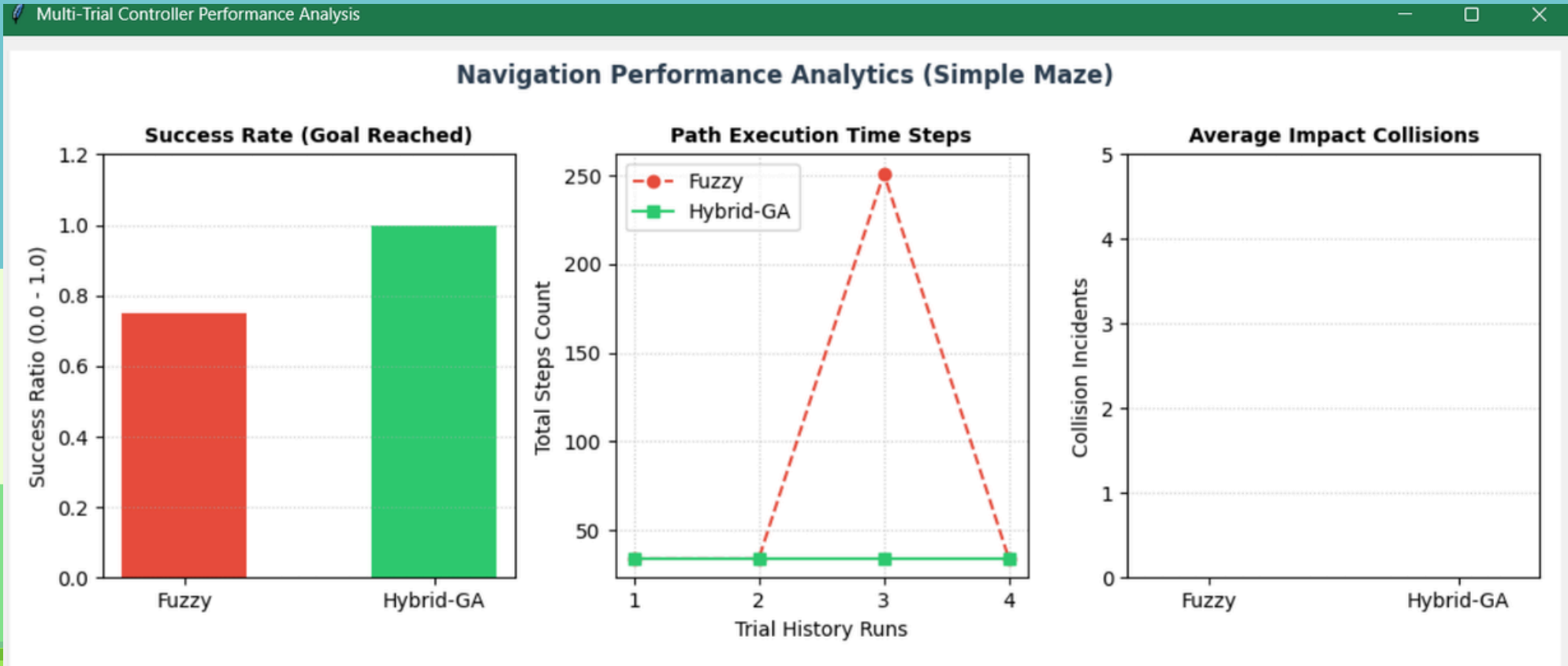
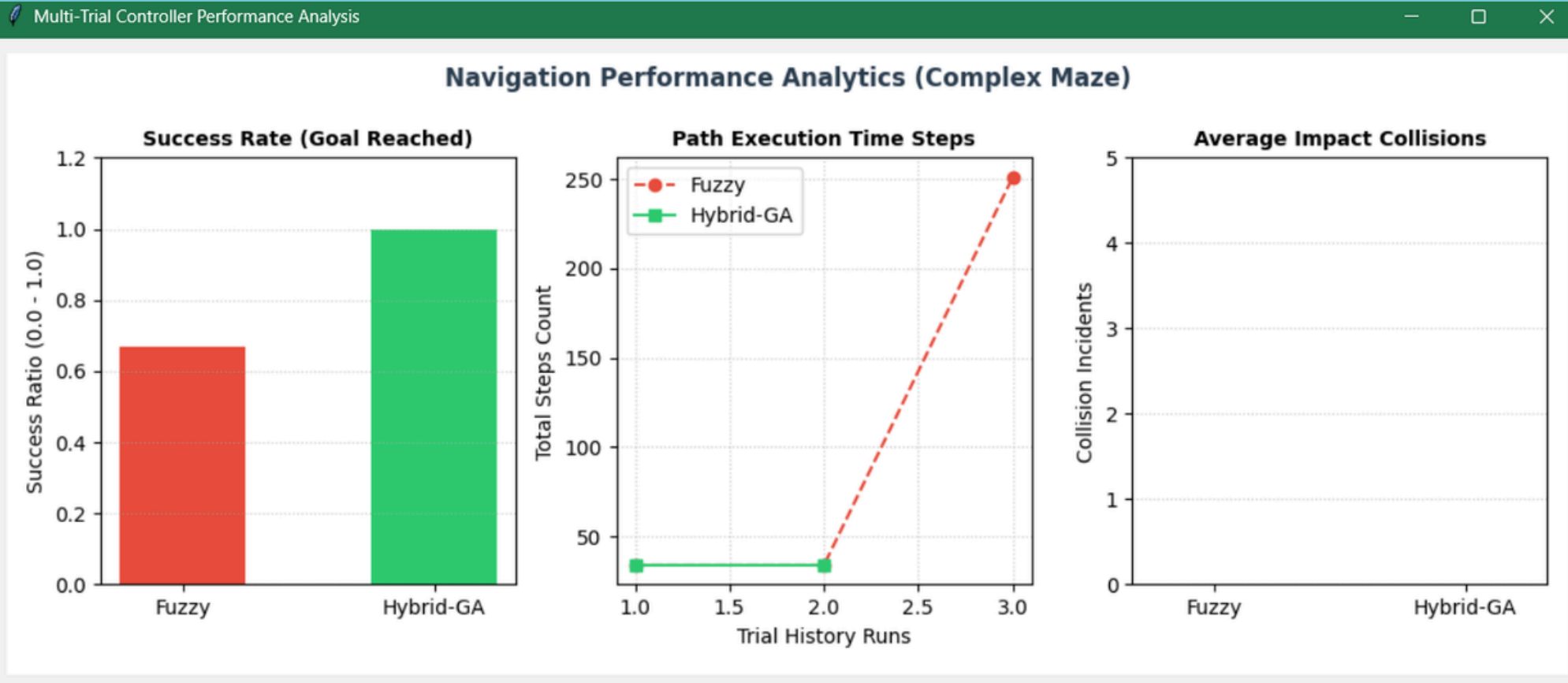




The controller was tested across at least two distinct environmental layouts.



Example of tested simulation in avoiding static obstacles showing that using Standard Fuzzy for Complex Maze was stuck



The generated graph of both Fuzzy and Hybrid-GA

PERFORMANCE EVALUATION RESULT

MCTA 3371

Performance Metric	Standard Fuzzy Controller	Hybrid Fuzzy-GA Controller
Goal Status	Goal reached	Goal reached
Time Steps	Low (approx. 30 to 40 steps)	Optimal (~25 to 35 steps)
Collisions	0 (Active avoidance)	0 (Perfect safety)
Goal Status	Stuck (local minimum)	Goal reached
Time Steps	Maxed Out (251 steps)	Optimal (~50 to 60 steps)
Collisions	0	0

DISCUSSION

MCTA 3371

- **Local Minima Discovery:** In the Complex Maze, the Standard Fuzzy approach fails because it only calculates optimal local choices, trapping itself in a physical loop.
- **Collision Safety:** Both controllers maintain 0 physical collisions because the collision filter completely restricts illegal entries into wall blocks.
- **Hybrid Dominance:** The Hybrid Fuzzy-GA resolves deadlocks by looking ahead globally, finding efficient, smooth path selections without wasting time steps.



CONCLUSION

MCTA 3371

This project successfully developed a soft-computing navigation framework within an interactive Graphical User Interface (GUI) environment. By integrating Fuzzy Logic for local reactive decision-making and a Genetic Algorithm (GA) for global path optimization, the system was able to navigate effectively through obstacle-filled environments. The results demonstrate that the combination of these two approaches provides a more efficient and reliable solution for autonomous navigation, highlighting the importance of hybrid intelligent techniques in solving complex path-planning problems.



CONCLUSION

MCTA 3371

Artificial Intelligence (AI) tools were used as a supporting resource during the development of this project, particularly for assisting with Tkinter Canvas layer management using `tag_raise()`, generating obstacle-grid layout patterns, and troubleshooting the integration of Matplotlib evaluation charts within Tkinter Toplevel windows. All AI-generated suggestions, code structures, and implementation approaches were carefully reviewed, tested, debugged, and modified before being incorporated into the final system. The project team takes full responsibility for the design, implementation, functionality, code correctness, and performance results presented in this simulation project.



The image is a pixelated screenshot of a 'Thank You' screen from the Super Mario Bros. game. The background is a solid light blue sky. In the center, the words 'THANK YOU' are written in a large, bold, orange-yellow pixel font with a thick black outline. Below this, the words 'FOR PLAYING!' are written in a smaller, white pixel font with a black outline. The bottom of the screen features a green grassy ground with a thin brown line representing the ground level. There are five green warp pipes: three floating in the sky (two on the left, one in the center) and two on the ground (one on the left, one on the right). The pipes have a green body with a lighter green vertical stripe and a black top rim. The overall style is classic 8-bit pixel art.

**THANK
YOU**

FOR PLAYING!